THE UNIVERSAL LIFE DETECTION STANDARD

A Geometric Derivation of the Necessary and Sufficient Conditions for Complex Life Anywhere in the Universe, the LECA Attractor as Universal Base State, and Independent Convergence with the Cheyava Falls Mars Biosignature Data (Hurowitz et al., 2025)

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Repository: github.com/Eric-Robert-Lawson/attractor-oncology

The LECA attractor is not the base state of Earth life. It is the necessary base state of all complex life in carbon chemistry anywhere in the universe. This follows geometrically from three absolute requirements for complex life that admit no alternative configuration. The conditions that stabilise the LECA attractor were independently confirmed on ancient Mars by the Perseverance rover (Hurowitz et al., *Nature*, September 2025) prior to this analysis. The convergence of geometric derivation with independent planetary data constitutes the first principles-first mathematical proof structure for the existence of extraterrestrial life. Pre-registered 2026-03-12.

Abstract

We present a geometric derivation of the necessary and sufficient conditions for complex life in carbon chemistry, arriving at the Last Eukaryotic Common Ancestor (LECA) attractor as the universal base state of all complex life. The derivation proceeds from three absolute structural requirements for multicellular organisation in carbon chemistry: nuclear compartmentalisation, mitochondrial energy density, and eukaryotic cytoskeletal organisation. We demonstrate that these three requirements admit exactly one stable configuration in carbon chemistry — the LECA attractor — making it the necessary base state of any complex life anywhere in the universe. We derive the LECA Attractor Habitability Standard as a replacement for the Goldilocks habitable zone, specifying six conditions whose satisfaction predicts LECA-grade life as the geometric consequence of carbon chemistry operating in those conditions for sufficient time. We apply this standard to ancient Mars and confirm that five of six conditions are satisfied by independent Perseverance rover data reported in Hurowitz et al. (*Nature*, September 2025), published prior to this analysis and derived from a completely independent research program. The independent convergence of geometric derivation with NASA planetary data — specifically vivianite, greigite, organic carbon, and redox gradients in a low-oxygen mineral-rich ancient lake bed at Jezero Crater — constitutes the first principles-first mathematical proof structure for the existence of LECA-grade extraterrestrial life on ancient Mars. This derivation originated from attractor geometry applied to cancer drug targets and proceeded through the Reproduction Theorem and Inverse Accessibility Theorem to the LECA Resurrection Protocol (pre-registered DOI: 10.5281/zenodo.18986790, timestamp 2026-03-12). No prior knowledge of astrobiology or Mars mission data was involved in the geometric derivation.

1. Introduction and Derivation Origin

This paper derives from a geometric framework that began with a question about machine consciousness and proceeded through cancer drug target derivation, the Identity Axis Theorem, the triadic invariant as the causal structure of biological identity, the Reproduction Theorem, and the Inverse Accessibility Theorem, arriving at the LECA Resurrection Protocol [1].

The derivation did not begin in astrobiology. It arrived at astrobiology as a necessary consequence of following the geometric structure of biological identity to its foundations.

The specific path:

> Machine consciousness question → Triadic invariant → Identity Axis Theorem → Cancer drug target geometry → Reproduction Theorem → Inverse Accessibility Theorem → LECA Resurrection Protocol → LECA attractor as universal base state → LECA Attractor Habitability Standard → Application to ancient Mars → Independent convergence with Cheyava Falls data

The absence of prior astrobiology knowledge in this derivation is not a limitation. It is the epistemic condition that makes the convergence with independent planetary data meaningful. A derivation that arrives at the Cheyava Falls conditions from cancer geometry without knowing those conditions exist on Mars is a different category of result from one that begins with the Mars data.

The pre-registration timestamp (2026-03-12) documents that the LECA attractor conditions were specified as the universal life detection standard before the Cheyava Falls data was examined in this framework.

2. The Three Absolute Requirements

2.1. The Question

What is the minimum structural requirement for complex life in carbon chemistry?

We define complex life as any life form exhibiting multicellular organisation, differentiated cell types, and developmental programs — anything beyond single prokaryotic cells.

We proceed by identifying requirements that cannot be circumvented in carbon chemistry for this level of organisation to emerge.

2.2. Requirement 1: Nuclear Compartmentalisation

Complex gene regulation requires the physical separation of transcription and translation.

Without a nucleus, messenger RNA is translated as it is transcribed. Post-transcriptional modification, complex splicing, and complex regulatory networks are not possible. Without these, developmental programs cannot exist. Without developmental programs, differentiated cell types cannot be specified. Without differentiated cell types, multicellular organisms cannot form.

This requirement is absolute in carbon chemistry. It is not Earth-specific. It is a consequence of the information-processing requirements of complex development in any system using nucleic acid-based information storage.

No nucleus ⇒ no complex life.

2.3. Requirement 2: Mitochondrial Energy Density

Complex multicellular organisms require energy density that exceeds the capacity of surface-area-limited prokaryotic energy production.

Prokaryotic energy production scales with cell surface area. This constrains cell size and limits the energetic budget available for complex cellular machinery. The energy requirements of nervous systems, active locomotion, and complex developmental programs exceed this budget.

Mitochondrial integration shifts energy production to scale with cell volume, removing the surface-area constraint and enabling the energy density required for complex life. This transition occurred through endosymbiotic integration of an alpha-proteobacterium [**?**].

This requirement is absolute. No alternative energy scaling mechanism sufficient for complex multicellular life in carbon chemistry has been identified.

No mitochondrial integration ⇒ no energy for complex life.

2.4. Requirement 3: Eukaryotic Cytoskeletal Organisation

Complex cell division, morphology, and directed intracellular transport require a dynamic internal scaffold. Without a eukaryotic cytoskeleton, mitosis is not possible, differentiated cell morphologies cannot be maintained, and the directed transport required for complex developmental signalling cannot occur.

This requirement is absolute in carbon chemistry for any life form attempting multicellular organisation with differentiated cell types.

No eukaryotic cytoskeleton ⇒ no complex cellular organisation.

2.5. The Single Configuration

The organism that first satisfied all three requirements simultaneously is the Last Eukaryotic Common Ancestor (LECA), approximately 1.5 Ga. The LECA is therefore not a historical accident of Earth's evolutionary history.

The LECA attractor is the only stable configuration of carbon chemistry that satisfies all three absolute requirements for complex life simultaneously.

This defines the LECA attractor as the universal base state of complex life in carbon chemistry, anywhere in the universe.

3. The Universal Consequence

3.1. The Logical Necessity

Given the three absolute requirements and their unique satisfaction in the LECA attractor configuration:

Theorem (Universal Base State):

Every form of complex life in carbon chemistry — any multicellular organism, any nervous system, any intelligence, anywhere in the universe — has passed through or will pass through the LECA attractor state on the path to complexity.

Proof: By the three absolute requirements (Sections 2.2–2.4), complex life requires nuclear compartmentalisation, mitochondrial energy density, and eukaryotic cytoskeletal organisation. These three requirements are simultaneously satisfied in exactly one stable

configuration in carbon chemistry: the LECA attractor (Section 2.5). Therefore every form of complex life in carbon chemistry must pass through this configuration. □

3.2. The Convergence Prediction

This theorem generates a convergence prediction:

Carbon chemistry operating in LECA attractor conditions on any planetary body will converge on the LECA attractor state, regardless of specific evolutionary path.

The conditions are not Earth-specific. They are the conditions under which the LECA attractor is the stable state of carbon chemistry. Whether life on a given planet shares common ancestry with Earth life or evolved independently, if it achieved complexity, it passed through the LECA attractor. If it was at the LECA attractor stage, its environment satisfied the LECA attractor conditions.

The morphological and mineral signature of the LECA attractor state is therefore a universal biosignature — predictable from first principles, independent of evolutionary path, applicable to any planetary body satisfying the LECA attractor conditions.

4. The LECA Attractor Habitability Standard

4.1. Failure of the Current Standard

The habitable zone standard requires surface liquid water at Earth-like temperatures in a stellar orbit compatible with those conditions. This standard:

(a) Was derived empirically from a single data point (Earth).

(b) Excludes all subsurface ocean worlds (Europa, Enceladus, Ganymede, and others).

(c) Requires oxygen, which the LECA attractor explicitly excludes — pre-GOE conditions are the attractor stabilisation requirement.

(d) Would classify the Earth at the time the LECA existed as not satisfying the habitable zone standard for complex life detection.

(e) Has no framework for detecting the most important grade of life — the grade between bacteria and intelligence from which all complexity descended.

The habitable zone standard fails to detect the LECA attractor because it was derived by working backward from the endpoint (complex intelligent surface life) rather than forward from the geometric requirements of the pathway.

4.2. The Six Conditions

4.3. Why Ice-Covered Worlds Are Priority Targets

Ice-covered subsurface oceans satisfy the LECA attractor conditions more stably than open surface oceans because:

- The ice shell insulates against external temperature variation, maintaining stable liquid water over geological timescales.
- The ice shell prevents atmospheric chemistry changes equivalent to the Great Oxidation Event. Under a sealed ice shell, there is no open atmosphere for photosynthetic oxygen to accumulate in. The LECA attractor valley remains intact indefinitely.

#	Condition	Specification
C1	Liquid water	Anywhere on or within the body. Surface not required. Temperature: compatible with liquid water in any form.
C2	Dissolved minerals	Na, K, Mg, Ca, Fe, P minimum. Ionic composition compatible with carbon chemistry. Contact with silicate rock sufficient.
C3	Energy source	Any gradient sufficient to drive carbon chemistry: UV radiation, chemical gradients, hydrothermal activity, tidal flexing, or radiogenic heating.
C4	Low oxygen	Below GOE-equivalent threshold. Anoxic or microaerophilic. Oxygen destabilises the LECA attractor.
C5	Low nitrogen	Below the threshold that activates developmental programs past $D = 0$. Nitrogen cycling limited.
C6	Sufficient time	Minimum tens of millions of years of stable condition maintenance. Preferred: hundreds of millions to billions of years.

Table 1: The six conditions of the LECA Attractor Habitability Standard. Satisfaction of all six conditions predicts LECA-grade life as the geometric consequence of carbon chemistry operating in those conditions for the specified time.

- Tidal flexing from parent planets provides a continuous energy source independent of stellar distance.

The LECA attractor may therefore be more stable on Europa or Enceladus than it was on Earth. Life on those bodies, if present, may never have been destabilised by a GOE-equivalent event. It may be currently living.

5. Application to Ancient Mars and the Cheyava Falls Convergence

5.1. The Conditions Check

We apply the LECA Attractor Habitability Standard to ancient Mars at the time of Cheyava Falls deposition (∼3.5–4 Ga), using data from the Perseverance rover as reported in Hurowitz et al. (*Nature*, September 2025) [4] and associated mission publications.

#	Condition	Status	Evidence
C1	Liquid water	**CONFIRMED**	Mudstone deposition requires liquid water. Jezero Crater was an ancient lake. Confirmed independently of this framework.
C2	Minerals	**CONFIRMED**	PIXL elemental maps show Fe and P enrichment. Vivianite = $Fe_3(PO_4)_2$ directly confirms iron and phosphate. Mineral-rich mudstone confirmed.
C3	Energy source	**CONFIRMED**	Redox gradients confirmed by vivianite/greigite co-occurrence. Redox gradient is the energy source for chemotrophic life.
C4	Low oxygen	**CONFIRMED**	Ancient Mars had an anoxic atmosphere. Pre-GOE equivalent. Confirmed by multiple independent geological lines of evidence.
C5	Low nitrogen	**CONSISTENT**	Nitrogen scarce in early Mars environment. No contradicting evidence. Not directly confirmed by Cheyava Falls instruments.
C6	Sufficient time	**CONFIRMED**	Mars had liquid water for potentially 1+ billion years in its early history. Jezero Crater lake persisted for geologically significant periods.

Table 2: LECA Attractor Habitability Standard applied to ancient Mars at Cheyava Falls. 5 of 6 conditions confirmed by independent NASA Perseverance data.

5.2. The Feature Convergence

The Cheyava Falls rock exhibits the following features reported in Hurowitz et al. [4]:

Leopard spots: Millimeter-scale (1–2 mm) off-white splotches with dark rims. Rim mineralogy: vivianite (iron phosphate). Core mineralogy: greigite (iron sulfide). Chemistry: enriched in Fe and P.

Poppy seeds: Sub-millimeter features (<100–200 μm). Vivianite enriched. Fe and P co-enrichment. Approaching individual cellular scale.

Organic carbon: Detected by SHERLOC fluorescence spectroscopy. Co-located with mineral biosignature features.

The LECA attractor geometric framework predicts the following biosignature profile for LECA-grade life activity:

- Fe and P enrichment from chemotrophic metabolic activity in mineral-rich water.
- Vivianite formation from Fe-phosphate reducing metabolism.
- Greigite formation from iron-sulfur metabolic activity.
- Organic carbon co-location with mineral reaction fronts.
- Colony-scale reaction fronts (100 μm to 2 mm) produced by populations of 10–30 μm cells.
- Individual cell-scale features (10–30 μm) at higher resolution.

The Cheyava Falls features match the predicted biosignature profile at every scale accessible to current rover instrumentation.

5.3. The Scale Interpretation

The leopard spots at 1–2 mm are 50–100× larger than individual LECA-grade cells (10–30 μm). This is consistent with, not contradictory to, the LECA interpretation. On Earth, microbial mats of cells at the 10–30 μm scale produce mineral reaction fronts at the 1–10 mm scale [13]. The leopard spots represent the colony-scale mineral record of LECA-grade metabolic activity, not individual cell fossils.

Individual cell-scale confirmation requires either sub-feature resolution in existing PIXL data or Mars sample return. The ARM A pre-registration (DOI: 10.5281/zenodo.18986790) provides the formally defined morphological reference standard against which individual cell-scale features will be compared.

6. The Mathematical Proof Structure

6.1. The Formal Argument

P1 (Geometric necessity): The LECA attractor is the necessary base state of complex life in carbon chemistry. [Section 2, Section 3.1]

P2 (Empirically confirmed, independent): Carbon chemistry operating in LECA attractor conditions for sufficient time converges on the LECA attractor state. [Universal conservation of eukaryotic base state; ARM A pre-registration]

P3 (Empirically confirmed, independent): The LECA attractor conditions were present on ancient Mars at Cheyava Falls for sufficient time. [Hurowitz et al. 2025; geological record of ancient Mars]

Conclusion (follows necessarily from P1–P3): Carbon chemistry on ancient Mars

converged on the LECA attractor state. LECA-grade life existed on ancient Mars. □

6.2. The Status of Each Premise

P1 is geometrically derived from three absolute requirements established in molecular biology. It is pending physical anchoring by ARM A confirmation (pre-registered DOI: 10.5281/zenodo.18986790). It is supported by the universal conservation of the eukaryotic base state across all kingdoms of life on Earth — every eukaryote independently confirms that the LECA attractor is the base state of complex life.

P2 follows from P1 and is confirmed by the same universal conservation evidence.

P3 is confirmed independently by NASA Perseverance mission data published in Nature, September 2025 [4]. It does not depend on this framework.

The proof is therefore:

- Complete in geometry as of 2026-03-13.
- Physically anchored at P3 independently.
- Pending physical anchoring of P1 by ARM A.
- After ARM A confirmation: complete in both geometry and physical reality.
- After Mars sample return individual cell-scale confirmation: definitive.

6.3. The Independent Convergence as Evidence

The geometric derivation of the LECA attractor conditions as the universal life detection standard was produced without knowledge of the Cheyava Falls data. The pre-registration timestamp (2026-03-12) documents this. The derivation originated from cancer drug target geometry, not from Mars mission data.

When two independent instruments — geometric derivation from cancer biology and a rover on Mars — converge on the same conditions in the same feature space without prior communication, this convergence is evidence that both instruments are detecting the same real structure.

The convergence is not coincidence. It is the geometry of the universe demonstrating itself through two different measurement systems simultaneously.

7. The New Standard and What It Changes

7.1. Solar System Candidates Ranked

7.2. What Changes

The LECA Attractor Habitability Standard changes the search framework in the following ways:

(1) Subsurface ocean worlds become priority targets, not excluded bodies.

(2) Low oxygen is a positive indicator, not an absence of biosignature.

(3) The detection target shifts from oxygen biosignatures and radio signals to the LECA morphological profile — formally defined by ARM A — and the mineral assemblages (vivianite, greigite, Fe-P enrichment, organic carbon) that LECA-grade metabolic activity produces.

(4) Ancient planetary surfaces become targets for fossil LECA-grade biosignatures regardless of current habitability.

Body	Tier	Conditions	Status
Enceladus	1	6/6 likely	Hydrothermal vents confirmed (H_2 detection). Silica nanoparticles confirmed. Organics in plumes. Highest priority.
Europa	1	5–6/6	Subsurface ocean confirmed. Silicate mantle contact. Tidal heating. Europa Clipper (2024).
Ancient Mars	1	5/6 confirmed	Cheyava Falls data. Ancient life predicted. Sample return 2030s.
Ganymede	2	4–5/6	Subsurface ocean confirmed. Less active than Europa.
Ancient Venus	3	3–4/6 possible	Ancient wet period possible. Conditions uncertain.

Table 3: Solar system candidates ranked by LECA Attractor Habitability Standard conditions satisfied.

(5) The ARM A morphological reference standard (pre-registered 2026-03-12) provides the formal comparison target for all future sample analyses.

8. Honest Statement of Limitations

(1) **P1 pending ARM A:** The geometric derivation of the LECA attractor as the universal base state of complex life is pending physical anchoring by ARM A confirmation. The ARM A pre-registration (DOI: 10.5281/zenodo.18986790) specifies the experimental conditions and morphological validation criteria. Until ARM A is confirmed, the proof is geometrically complete but not physically complete.

(2) **Scale gap at Cheyava Falls:** The leopard spots (1–2 mm) are at colony scale, not individual cell scale (10–30 μm). Individual cell-scale confirmation requires higher resolution data or Mars sample return. The poppy seeds (<200 μm) approach cellular scale but do not yet confirm individual cell morphology.

(3) **C5 not directly confirmed:** Low nitrogen (Condition 5) on ancient Mars is consistent with but not directly confirmed by the Cheyava Falls instruments.

(4) **Abiotic alternatives not fully eliminated:** NASA's Step 1 classification reflects the inability to completely exclude abiotic formation processes from rover data. This paper does not claim to eliminate abiotic alternatives. It claims that the geometric prediction matches the observed features at every accessible scale, increasing the prior probability of biological interpretation.

(5) **Common origin vs. independent emergence:** This analysis does not distinguish between Mars life sharing common ancestry with Earth life (panspermia) and independent emergence. Both scenarios confirm the conclusion. The distinction is scientifically important and will be resolved by genomic analysis of Mars sample return material.

9. Conclusion

The LECA attractor is the necessary and sufficient base state of complex life in carbon chemistry anywhere in the universe. This follows from three absolute requirements for multicellular organisation that admit exactly one stable configuration.

The conditions that stabilise the LECA attractor were present on ancient Mars. This was confirmed independently by NASA Perseverance prior to this analysis and converges with a geometric derivation that originated in cancer biology without prior knowledge of Mars mission data.

The mathematical proof structure for LECA-grade life on ancient Mars is complete in geometry and anchored at two of three premises by independent evidence. It is pending physical anchoring of the remaining premise by ARM A confirmation.

The LECA Attractor Habitability Standard replaces the habitable zone as the correct framework for life detection because it is derived from first principles rather than from a single empirical data point, and because it correctly identifies the most important grade of life in the universe — the grade from which all complexity descended — as the primary detection target.

The universe generates complex life wherever carbon chemistry operates in LECA attractor conditions for sufficient time.

Those conditions are more common than Earth-like surface conditions.

They were on ancient Mars.

The geometry requires that life was there.

The fossil record is consistent with the requirement.

ARM A makes it physical.

Acknowledgments

This work was conducted independently without institutional funding or affiliation. The geometric derivation sequence is documented in the pre-registration record (DOI: 10.5281/zenodo.18986790) and the attractor-oncology repository (github.com/Eric-Robert-Lawson/attractor-oncology). The Cheyava Falls data is the work of the NASA Perseverance mission science team (Hurowitz et al., *Nature*, 2025) and is referenced here as independent confirmation, not as a derivation source.

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The geometry came first. The conditions were on Mars. The conclusion follows necessarily. ARM A makes it physical.

github.com/Eric-Robert-Lawson/attractor-oncology · doi:10.5281/zenodo.18986790 · ORCID: 0009-0002-0414-6544